

Classes And Objects

A class is used for binding data and the functions that work on them. In other words, a class may also be considered as an expanded form of a data structure. It has the unique feature of holding data and functions together. While creating a class, we use the keyword `class`. Classes can also be instantiated and the object is an instantiation of the class. While declaring objects, assume `class` as the user-defined data type, and `object` as the variable.

A class is declared as follows:

```
class class_name {  
    access_specifier_1:  
    member_1;  
    access_specifier_2:  
    member_2;  
} object_names;
```

Here, `class_name` is the name of the particular class or a valid identifier of it. Access specifiers determine the visibility of the members. An access specifier can be either `private`, `protected` or `public`. A member when declared as `private` can be accessible only by the other members of the same class, while a member declared as `protected` can be accessed by members of the same class, as well as members from their derived classes. Similarly, if we declare a member as `public`, it can be accessed from any part of the program where it is visible.

However, by default all data members of a class that are declared are `private` in nature. `member_1`, `member_2`, etc. represent data members or methods of the class declared. `Object_names` represents the name of the object. If we want to create multiple objects, then their names should be separated by commas and a semicolon, else the declaration is termi-